

Problems and Solutions - Part 1

numerical differentiation
numerical time integration of initial-value problems
interpolation

numerical differentiation

Concepts and definitions you need to know (multiple-choice questions to test understanding)

- Landau's $\mathcal{O}$ order symbol (arithmetic operations)
- Taylor series with remainder
- Intermediate-Value Theorem
- finite-difference approximation of a derivative
- truncation error
- order of approximation
- general method for deriving difference formulas
- Richardson's extrapolation method

Numerical/derivation questions you may get

- Derive a finite-difference formula that approximates a derivative up to a given order
- Compute the truncation error of a given finite-difference approximation
- Use Richardson's method to estimate the order of approximation or extrapolate a finite-difference formula to a higher-order formula

Example 1 (derivation of a finite-difference formula)

Question. Derive an $\mathcal{O}(h^2)$ difference formula for the first derivative $f'(x)$ using the nodes $x - 2h$, x , $x + h$.

Solution. The difference formula $Q(h)$ for $f'(x)$ for the given nodes has the form:

$$Q(h) = \frac{1}{h}(\alpha_{-2}f(x - 2h) + \alpha_0f(x) + \alpha_1f(x + h))$$

To find α_{-2} , α_0 and α_1 we use the general method. We need two Taylor series with $\mathcal{O}(h^3)$ remainders:

$$f(x - 2h) = f(x) - 2hf'(x) + 2h^2f''(x) + \mathcal{O}(h^3)$$

$$f(x + h) = f(x) + hf'(x) + \frac{h^2}{2}f''(x) + \mathcal{O}(h^3)$$

Example 1

$$\begin{aligned}Q(h) &= \frac{1}{h}(\alpha_{-2}f(x-2h) + \alpha_0f(x) + \alpha_1f(x+h)) \\&= \frac{1}{h}\alpha_{-2}(f(x) - 2hf'(x) + 2h^2f''(x) + \mathcal{O}(h^3)) \\&\quad + \frac{1}{h}\alpha_0f(x) \\&\quad + \frac{1}{h}\alpha_1\left(f(x) + hf'(x) + \frac{h^2}{2}f''(x) + \mathcal{O}(h^3)\right) \\&= \frac{1}{h}(\alpha_{-2} + \alpha_0 + \alpha_1)f(x) \\&\quad + \frac{1}{h}(-2h\alpha_{-2} + h\alpha_1)f'(x) \\&\quad + \frac{1}{h}\left(2h^2\alpha_{-2} + \frac{h^2}{2}\alpha_1\right)f''(x) \\&\quad + \frac{1}{h}(\alpha_{-2}\mathcal{O}(h^3) + \alpha_1\mathcal{O}(h^3))\end{aligned}$$

Example 1

$$\begin{aligned}Q(h) &= \frac{1}{h}(\alpha_{-2} + \alpha_0 + \alpha_1)f(x) \\&\quad + \frac{1}{h}(-2h\alpha_{-2} + h\alpha_1)f'(x) \\&\quad + \frac{1}{h}\left(2h^2\alpha_{-2} + \frac{h^2}{2}\alpha_1\right)f''(x) \\&\quad + \mathcal{O}(h^2) \\&= f'(x) + \mathcal{O}(h^2)\end{aligned}$$

$$\begin{cases} \frac{1}{h}(\alpha_{-2} + \alpha_0 + \alpha_1) = 0 \\ \frac{1}{h}(-2h\alpha_{-2} + h\alpha_1) = 1 \\ \frac{1}{h}\left(2h^2\alpha_{-2} + \frac{h^2}{2}\alpha_1\right) = 0 \end{cases} \Rightarrow \begin{cases} \alpha_{-2} + \alpha_0 + \alpha_1 = 0 \\ -2\alpha_{-2} + \alpha_1 = 1 \\ 4\alpha_{-2} + \alpha_1 = 0 \end{cases}$$

Example 1

From the system

$$\begin{cases} (1)\alpha_{-2} + (1)\alpha_0 + (1)\alpha_1 = 0 \\ (-2)\alpha_{-2} + (0)\alpha_0 + (1)\alpha_1 = 1 \\ (4)\alpha_{-2} + (0)\alpha_0 + (1)\alpha_1 = 0 \end{cases}$$

we deduce the

Augmented matrix of the system

$$\begin{bmatrix} 1 & 1 & 1 & 0 \\ -2 & 0 & 1 & 1 \\ 4 & 0 & 1 & 0 \end{bmatrix}$$

Example 1

Solving the system of equations by elimination:

$$\begin{cases} \alpha_{-2} + \alpha_0 - 4\alpha_{-2} = 0 \\ -2\alpha_{-2} - 4\alpha_{-2} = 1 \\ \alpha_1 = -4\alpha_{-2} \end{cases} \Rightarrow \begin{cases} \alpha_0 = 3\alpha_{-2} \\ -6\alpha_{-2} = 1 \\ \alpha_1 = -4\alpha_{-2} \end{cases}$$
$$\begin{cases} \alpha_0 = -\frac{1}{2} \\ \alpha_{-2} = -\frac{1}{6} \\ \alpha_1 = \frac{2}{3} \end{cases} \Rightarrow \begin{bmatrix} \alpha_{-2} \\ \alpha_0 \\ \alpha_1 \end{bmatrix} = \begin{bmatrix} -\frac{1}{6} \\ -\frac{1}{2} \\ \frac{2}{3} \end{bmatrix}$$

Coefficients of the $\mathcal{O}(h^2)$ finite-difference formula for $f'(x)$

$$\begin{bmatrix} \alpha_{-2} \\ \alpha_0 \\ \alpha_1 \end{bmatrix} = \begin{bmatrix} -\frac{1}{6} \\ -\frac{1}{2} \\ \frac{2}{3} \end{bmatrix}$$

Example 2 (truncation error of a FD formula)

Question. Determine the truncation error of the following $\mathcal{O}(h^2)$ difference formula approximating $f'(x)$:

$$Q(h) = \frac{1}{h} \left(-\frac{1}{6}f(x-2h) - \frac{1}{2}f(x) + \frac{2}{3}f(x+h) \right)$$

Solution. We need two Taylor series with explicit remainders:

$$f(x-2h) = B_0f(x) + B_1hf'(x) + B_2h^2f''(x) + B_3f^{(3)}(\xi_-)h^3, \quad \xi_- \in [a_-, b_-]$$

$$f(x+h) = C_0f(x) + C_1hf'(x) + C_2h^2f''(x) + C_3f^{(3)}(\xi_+)h^3, \quad \xi_+ \in [a_+, b_+]$$

Answer: correct form of the Taylor series

$$f(x-2h) = f(x) - 2hf'(x) + 2h^2f''(x) - \frac{8}{6}f^{(3)}(\xi_-)h^3, \quad \xi_- \in [x-2h, x]$$

$$f(x+h) = f(x) + hf'(x) + \frac{h^2}{2}f''(x) + \frac{1}{6}f^{(3)}(\xi_+)h^3, \quad \xi_+ \in [x, x+h]$$

Example 2

Truncation error is defined as:

$$|f'(x) - Q(h)|$$

$$\begin{aligned} |f'(x) - Q(h)| &= \left| f'(x) - \frac{1}{h} \left(-\frac{1}{6}f(x-2h) - \frac{1}{2}f(x) + \frac{2}{3}f(x+h) \right) \right| \\ &= \left| f'(x) + \frac{1}{6h} \left(f(x) - 2hf'(x) + 2h^2f''(x) - \frac{8}{6}f^{(3)}(\xi_-)h^3 \right) \right. \\ &\quad \left. + \frac{1}{2h}f(x) \right. \\ &\quad \left. - \frac{2}{3h} \left(f(x) + hf'(x) + \frac{h^2}{2}f''(x) + \frac{1}{6}f^{(3)}(\xi_+)h^3 \right) \right| \\ &= \left| \left(\frac{1}{6h} + \frac{1}{2h} - \frac{2}{3h} \right) f(x) + \left(1 - \frac{2}{6} - \frac{2}{3} \right) f'(x) \right. \\ &\quad \left. + \left(\frac{2h^2}{6h} - \frac{2h^2}{(3h)2} \right) f''(x) \right. \\ &\quad \left. + \left(-\frac{8h^3}{(6h)6}f^{(3)}(\xi_-) - \frac{2h^3}{(3h)6}f^{(3)}(\xi_+) \right) \right| \end{aligned}$$

Example 2

Answer with ξ_- and ξ_+

$$|f'(x) - Q(h)| = \left| \frac{2}{9}f^{(3)}(\xi_-) + \frac{1}{9}f^{(3)}(\xi_+) \right| h^2$$
$$\xi_- \in [x - 2h, x], \quad \xi_+ \in [x, x + h]$$

Example 2

To obtain the answer with one ξ , use the Intermediate Value Theorem (will be provided):

Theorem: Intermediate Value

Let $f \in C[a, b]$ and $f(a) \neq f(b)$. For any number F between $f(a)$ and $f(b)$ there exists a number $c \in (a, b)$ such that $f(c) = F$.

Example 2

In these formulas we often have the expression of the form $c_1 f^{(n)}(\xi_-) + c_2 f^{(n)}(\xi_+)$. You can use the following Lemma (will be provided).

Lemma: Range Interpolation

Let $f \in C[a, b]$ and $f(a) \neq f(b)$. Then, the value $F = \alpha f(a) + (1 - \alpha)f(b)$, where $0 < \alpha < 1$, is between $f(a)$ and $f(b)$.

Proof: If $f(a) < f(b)$ and $0 < \alpha < 1$, then:

$$\begin{aligned} f(a) &< \alpha f(a) + (1 - \alpha)f(b) < f(b) \\ (1 - \alpha)f(a) &< (1 - \alpha)f(b) < f(b) - \alpha f(a) \\ -\alpha f(b) &< -\alpha f(a) \\ f(b) &> f(a) \end{aligned}$$

Similarly for $f(a) > f(b)$. In summary: $\alpha f(a) + (1 - \alpha)f(b)$ with $0 < \alpha < 1$ is always between $f(a)$ and $f(b)$. □

Example 2

How to use this Lemma? We want to rewrite the result with ξ_- and ξ_+ as follows:

$$|f'(x) - Q(h)| = \left| \frac{2}{9}f^{(3)}(\xi_-) + \frac{1}{9}f^{(3)}(\xi_+) \right| h^2 \leq \beta \left| f^{(3)}(\xi) \right| h^2$$
$$\xi_- \in [x - 2h, x], \quad \xi_+ \in [x, x + h], \quad \xi \in [x - 2h, x + h]$$

Since one of the multipliers could have also been negative, first we transform the result with ξ_- and ξ_+ to the following form:

$$|f'(x) - Q(h)| = \left| c_1 f^{(3)}(\xi_-) + c_2 f^{(3)}(\xi_+) \right| h^2 \leq \left(|c_1| \left| f^{(3)}(\xi_-) \right| + |c_2| \left| f^{(3)}(\xi_+) \right| \right) h^2$$

Now, we look for β and $0 < \alpha < 1$ such that

$$|c_1| \left| f^{(3)}(\xi_-) \right| + |c_2| \left| f^{(3)}(\xi_+) \right| = \beta \left(\alpha \left| f^{(3)}(\xi_-) \right| + (1 - \alpha) \left| f^{(3)}(\xi_+) \right| \right)$$

$$\begin{cases} |c_1| = \beta\alpha \\ |c_2| = \beta(1 - \alpha) \end{cases} \Rightarrow \begin{cases} |c_1| = \beta\alpha \\ |c_2| = \beta - |c_1| \end{cases} \Rightarrow \begin{cases} \alpha = \frac{|c_1|}{|c_1| + |c_2|} \\ \beta = |c_1| + |c_2| \end{cases}$$

Example 2

In the present case, $c_1 = 2/9$ and $c_2 = 1/9$. Hence, $\beta = |c_1| + |c_2| = 1/3$, and

Answer with single ξ

$$|f'(x) - Q(h)| \leq \frac{1}{3} \left| f^{(3)}(\xi) \right| h^2, \quad \xi \in [x - 2h, x + h]$$

Example 3 (Richardson extrapolation)

Question. Let $Q(h)$ be the numerical approximation of M .

- 1 Assuming that values $Q(kh)$, $Q(mh)$ and $Q(nh)$ are known for some $k, m, n > 0$ and $k \neq m \neq n$, estimate the order of the approximation $Q(h)$ (in steps)
- 2 Given the formula $Q(h)$, derive the formula $\tilde{Q}(h)$ for a higher-order approximation

Solution. See Lectures for the logic. Eliminate M and C and find p from the relation $M - Q(h) = Ch^p$. To derive $\tilde{Q}(h)$, use the main assumption of the Richardson extrapolation method:

$$\begin{aligned}M - Q(h) &= Ch^p + \mathcal{O}(h^{p+1}), \\M - Q(h) - Ch^p &= \mathcal{O}(h^{p+1}),\end{aligned}$$

i.e., $\tilde{Q}(h) = Q(h) + Ch^p$. Assuming that the order p is known, here you will need to eliminate M and find C from the relation $M - Q(h) = Ch^p$.

numerical time integration of initial-value problems

Concepts and definitions you need to know (multiple-choice questions to test understanding)

For both a single equation and a system of equations:

- initial-value problem
- perturbed problem, initial value problem for the perturbation
- stability and absolute stability of problems with $f(y, t) = \lambda y + g$ and $\mathbf{f}(\mathbf{y}, t) = A\mathbf{y} + \mathbf{g}$
- idea of numerical time integration
- Forward-Euler, Backward-Euler, Trapezoidal, Modified-Euler
- amplification factor/matrix of a time integration method
- stability and absolute stability of a time integration method
- local and global truncation errors of a method
- consistency, convergence, Lax's Theorem

Questions you may get

- determine the amplification factor/matrix of a given time integration method
- derive the stability condition for a given method
- determine the order of the local and global truncation errors of a given method

Example 1 (amplification matrix)

Question. Determine the amplification matrix of the Modified-Euler Method for systems:

$$\begin{aligned}\bar{\mathbf{w}}_{n+1} &= \mathbf{w}_n + \Delta t \mathbf{f}(t_n, \mathbf{w}_n) \\ \mathbf{w}_{n+1} &= \mathbf{w}_n + \frac{\Delta t}{2} \mathbf{f}(t_n, \mathbf{w}_n) + \frac{\Delta t}{2} \mathbf{f}(t_n, \bar{\mathbf{w}}_{n+1})\end{aligned}$$

Solution. On paper you use the following notation for vectors:

$$\begin{aligned}\bar{\underline{\mathbf{w}}}_{n+1} &= \underline{\mathbf{w}}_n + \Delta t \underline{\mathbf{f}}(t_n, \underline{\mathbf{w}}_n) \\ \underline{\mathbf{w}}_{n+1} &= \underline{\mathbf{w}}_n + \frac{\Delta t}{2} \underline{\mathbf{f}}(t_n, \underline{\mathbf{w}}_n) + \frac{\Delta t}{2} \underline{\mathbf{f}}(t_n, \bar{\underline{\mathbf{w}}}_{n+1})\end{aligned}$$

Amplification factors are always derived for the test problem:

$$\underline{\epsilon}' = A\underline{\epsilon}, \quad \underline{\epsilon}(0) = \underline{\epsilon}_0,$$

where A is the Jacobian matrix of $\underline{\mathbf{f}}(t, \underline{\mathbf{y}})$ at some $\hat{t}$ and $\hat{\underline{\mathbf{y}}}$. You can use the general notation A for this matrix, unless additional information about $\underline{\mathbf{f}}$ is provided.

Example 1

Applying the Modified-Euler Method to the test problem we get:

$$\underline{\bar{w}}_{n+1} = \underline{w}_n + \Delta t A \underline{w}_n$$

$$\underline{w}_{n+1} = \underline{w}_n + \frac{\Delta t}{2} A \underline{w}_n + \frac{\Delta t}{2} A \underline{\bar{w}}_{n+1}$$

After substitution we get:

$$\underline{w}_{n+1} = \underline{w}_n + \frac{\Delta t}{2} A \underline{w}_n + \frac{\Delta t}{2} A (\underline{w}_n + \Delta t A \underline{w}_n)$$

Bring it to the form $\underline{w}_{n+1} = (\cdots) \underline{w}_n$

$$\underline{w}_{n+1} = \left(I + \frac{\Delta t}{2} A + \frac{\Delta t}{2} A + \frac{(\Delta t)^2}{2} A^2 \right) \underline{w}_n$$

The thing inside the brackets is the amplification matrix:

Answer: amplification matrix

$$Q = I + \Delta t A + \frac{(\Delta t)^2}{2} A^2$$

Example 2 (stability condition)

Question. Derive the absolute stability condition of the Modified-Euler Method for systems:

$$\begin{aligned}\bar{\mathbf{w}}_{n+1} &= \mathbf{w}_n + \Delta t \mathbf{f}(t_n, \mathbf{w}_n) \\ \mathbf{w}_{n+1} &= \mathbf{w}_n + \frac{\Delta t}{2} \mathbf{f}(t_n, \mathbf{w}_n) + \frac{\Delta t}{2} \mathbf{f}(t_n, \bar{\mathbf{w}}_{n+1})\end{aligned}$$

applied to the problem:

$$\mathbf{y}' = A\mathbf{y}, \quad \mathbf{y}(0) = \mathbf{y}_0,$$

where the matrix $A \in \mathbb{R}^{n \times n}$ has the eigenvalues with the properties $\operatorname{Re}(\lambda_j) < 0$, $\operatorname{Im}(\lambda_j) = 0$, $j = 1, \dots, n$.

Solution. You should remember that a time integration method applied to a problem is stable if:

- 1 The problem is stable. In the linear/test problem case the problem is stable if the eigenvalues λ_j of the matrix A have the property $\operatorname{Re}(\lambda_j) < 0$.
- 2 The method is absolutely stable, i.e., the eigenvalues $\tilde{\lambda}_j$ of the amplification matrix Q of the method have the property $|\tilde{\lambda}_j| < 1$ for all $j = 1, \dots, n$.

Example 2

- (1) Observe that the problem is stable.
- (2) Apply the method to the problem to get the amplification matrix (see previous example)

$$Q = I + \Delta t A + \frac{(\Delta t)^2}{2} A^2$$

Express the eigenvalues $\tilde{\lambda}_j$ of Q via the eigenvalues λ_j of A . To do this, let $\underline{v}_j$ be the eigenvector of A , i.e., $A\underline{v}_j = \lambda_j \underline{v}_j$. Then, noting that $A^2 \underline{v}_j = A(\lambda_j \underline{v}_j) = \lambda_j A \underline{v}_j = \lambda_j^2 \underline{v}_j$, we get

$$\begin{aligned} Q \underline{v}_j &= \left(I + \Delta t A + \frac{(\Delta t)^2}{2} A^2 \right) \underline{v}_j = I \underline{v}_j + \Delta t A \underline{v}_j + \frac{(\Delta t)^2}{2} A^2 \underline{v}_j \\ &= \underline{v}_j + \Delta t \lambda_j \underline{v}_j + \frac{(\Delta t)^2}{2} \lambda_j^2 \underline{v}_j = \left(1 + \Delta t \lambda_j + \frac{(\Delta t)^2}{2} \lambda_j^2 \right) \underline{v}_j = \tilde{\lambda}_j \underline{v}_j \end{aligned}$$

Hence,

$$\tilde{\lambda}_j = 1 + \Delta t \lambda_j + \frac{(\Delta t)^2}{2} \lambda_j^2$$

and $\text{Im}(\tilde{\lambda}_j) = 0$, since λ_j are real.

Example 2

Answer: implicit stability condition

$$\left| 1 + \Delta t \lambda_j + \frac{(\Delta t)^2}{2} \lambda_j^2 \right| < 1, \quad \text{for all } j = 1, \dots, n$$

To find a more explicit stability condition, we note that λ_j are real and $\lambda_j = -|\lambda_j|$. Thus,

$$-1 < 1 - \Delta t |\lambda_j| + \frac{(\Delta t)^2}{2} |\lambda_j|^2 < 1$$

$$-2 < -\Delta t |\lambda_j| + \frac{(\Delta t)^2}{2} |\lambda_j|^2 < 0$$

$$-2 < \Delta t |\lambda_j| \left(-1 + \frac{\Delta t}{2} |\lambda_j| \right) < 0$$

Example 2

Let $\Delta t > 0$. Then, for the right inequality to hold we need to have:

Answer: explicit stability condition

$$0 < \Delta t < \frac{2}{|\lambda_j|}, \quad \text{for all } j = 1, \dots, n$$

Example 2

If you are curious about the left inequality:

$$\begin{aligned}-1 &< 1 - \Delta t |\lambda_j| + \frac{(\Delta t)^2}{2} |\lambda_j|^2 < 1 \\ 0 &< 2 - |\lambda_j| \Delta t + \frac{|\lambda_j|^2}{2} (\Delta t)^2\end{aligned}$$

This is a parabola with Δt playing the role of x . Its "horns" point upwards. Hence, we satisfy the condition if this parabola does not cross the " x " axis, i.e., when the equation:

$$2 - |\lambda_j| \Delta t + \frac{|\lambda_j|^2}{2} (\Delta t)^2 = 0$$

has no real roots. The discriminant $D = b^2 - 4ac$ must then satisfy $D < 0$:

$$\begin{aligned}D &= |\lambda_j|^2 - 4(2) \frac{|\lambda_j|^2}{2} < 0 \\ &= -3|\lambda_j|^2 < 0\end{aligned}$$

which is always true.

Example 3 (local truncation error)

Question. Determine the order of the local truncation error of the Modified-Euler Method:

$$\begin{aligned}\bar{w}_{n+1} &= w_n + \Delta t f(t_n, w_n) \\ w_{n+1} &= w_n + \frac{\Delta t}{2} f(t_n, w_n) + \frac{\Delta t}{2} f(t_{n+1}, \bar{w}_{n+1})\end{aligned}$$

Solution. See Lecture 5. You need to remember the definition of the local truncation error:

$$\tau_{n+1} = \frac{y_{n+1} - z_{n+1}}{\Delta t}$$

and of z_{n+1} , which is obtained by letting $w_n = y_n$ and applying the method in question to get z_{n+1} as:

$$\begin{aligned}\bar{z}_{n+1} &= y_n + \Delta t f(t_n, y_n) \\ z_{n+1} &= y_n + \frac{\Delta t}{2} f(t_n, y_n) + \frac{\Delta t}{2} f(t_{n+1}, \bar{z}_{n+1})\end{aligned}$$

Example 3

Answer: local truncation error

$$\tau_{n+1} = \mathcal{O}((\Delta t)^2)$$

interpolation

Concepts and definitions you need to know

- linear and polynomial functions
- Lagrange form of linear interpolation
- error of linear interpolation
- Lagrange form of polynomial interpolation
- Lagrange basis functions
- error of polynomial approximation
- Runge phenomenon and Chebyshev nodes (basic idea)
- idea of splines, natural cubic spline (definition)

Questions you may get

- what is the relation between the number of nodes and the degree of interpolating polynomials?
- what is the order of the error when a function is approximated by its interpolating polynomial?
- make and justify the choice between interpolation methods for given function and nodes
- given a function $f(x)$ and two points a and b , construct the linear interpolation $L_1(x)$
- write down the Lagrange form of interpolating polynomial $L_n(x)$ of a given degree n
- write down the Lagrange basis function $L_{k,n}(x)$ of a given degree n in a given point x_k